Assessment Matrix for MSc Individual Projects

(Academic Year 2024-25 onwards)

1. Quality of Technical Work and Continuous Assessment of Student Performance (worth 25%)

(To be completed by project or placement supervisor only)

Grade Range (Highest to Lowest)	**A1, A2, A3, A4, A5**	**B1, B2, B3**	**C1, C2, C3**	**D1, D2, D3**	**E1, E2, E3**	**F1, F2, F3**	**G1, G2, H**
Descriptor	**Excellent**	**Very Good**	**Good**	**Satisfactory**	**Weak**	**Poor**	**G: Very Poor** **H: No Attainment**
Planning *(Weighting = 1)*	*Professional planning, managing a complex problem, mitigating risks, addressing sustainability and reserving time to be inclusive and ethical.*	*Very good planning, addressing a complex problem, mitigating risks, addressing sustainability and reserving time to be inclusive and ethical.*	*Good planning, addressing a complex problem, mitigating risks, considering sustainability and reserving time to be inclusive and ethical.*	*Basic planning leading to a satisfactory solution of the problem. Little consideration of risk mitigation and sustainability. Little time invested to be inclusive and ethical.*	*Weak planning leading only to a dubious solution of the problem, no consideration of risk mitigation or sustainability. No time invested to be inclusive and ethical.*	*Poor planning leading to a poor or no solution of the problem. No consideration of risk mitigation or sustainability. No time invested to be inclusive and ethical.*	*Little or no evidence of any planning and no solution presented.*
Initiative *(Weighting = 1)*	*Major input to the technical work; took ownership of both the technical content and overall project direction. Worked consistently during the project duration.*	*Very good input to the technical work; drove the project forward. Worked consistently during the whole project.*	*Good input to the project with some help from the supervisor to overcome specific technical problems. Worked mostly during the whole project duration.*	*Little initiative and relied on the supervisor to move the project forward. Worked only intermittently during the project duration.*	*Weak initiative and the supervisor had to drive it instead. Worked only for short periods during the project duration.*	*No initiative. Ignored warnings, reminders and advice by the supervisor. Worked only for short periods during the project duration.*	*Student contributed very little or nothing to the project.*
Professional Conduct *(Weighting =1)*	*Student integrated fully into the engineering environment, contributed widely as a valued peer, and espoused broader ethical values in the project work.*	*Student worked very well within the engineering environment (with rare difficulties) and understood broader ethical project aspects.*	*Interaction with colleagues, where appropriate, was good. Practiced professional behaviour and noted broader ethical aspects.*	*Interaction with colleagues, where appropriate, was satisfactory. There was evidence of understanding professional behaviour.*	*Student did not integrate into the engineering environment and had difficulty in operating with colleagues on a day-to-day basis.*	*The student found operating professionally to be challenging, with respect to colleagues and to wider ethical and safety aspects of the project*	*The student did not operate at any meaningful level in a professional environment*
Technical Quality of Work *(Weighting = 2)*	*Excellent work of publishable quality and insight. A rigorous treatment of a limited technical or creative design problem with excellent analysis or creative design choices.*	*Very good quality work with only minor failings, and clear insights and judgement for a complex multi-faceted engineering problem.*	*Competent work, trustworthy results, and a good level of insight into a complex engineering problem supported by suitable analyses.*	*Satisfactory solution of a limited technical or creative design problem with satisfactory analysis or creative design choices; some deficiencies in understanding.*	*Some work of limited technical quality, but only in one aspect of a complex problem. Attainment at a very modest level.*	*Very little evidence of master's level work and results technically dubious.*	*No output of any value.*

Assessment Matrix for MSc Individual Projects
(Academic Year 2024-25 onwards)

2. Dissertation (worth 60%)

(To be completed by James Watt School of Engineering staff only.)

Grade Range (Highest to Lowest)	**A1, A2, A3, A4, A5**	**B1, B2, B3**	**C1, C2, C3**	**D1, D2, D3**	**E1, E2, E3**	**F1, F2, F3**	**G1, G2, H**
Descriptor	**Excellent**	**Very Good**	**Good**	**Satisfactory**	**Weak**	**Poor**	**G: Very Poor** **H: No Attainment**
Presentation *(Weighting = 1)*	*Professional presentation. Figures created by student enlighten and are integral to the narrative flow.*	*Clear and consistent presentation which is easy to read. Most figures are clear and well-presented and customised to establish the narrative.*	*Minor flaws in the presentation and clarity of the figures. Typically, some figures from web, which are not tailored to the narrative. Result figures have minor issues.*	*Basic presentational errors. Figures often sourced from web and do not support the narrative. Some result figures are hard to read, and/or some are not driving the narrative.*	*Significant flaws in the presentation. Most figures from web with loose connection to main text. Most results are of poor quality and do not drive the narrative or relate to it.*	*A substantial proportion of figures from web that are unrelated to the narrative. Results cannot be understood due to poor labelling / captioning.*	*A messy report. Figures do not match the narrative and results are unclear.*
Problem & context *(Weighting = 2)*	*Professional evaluation of the state of the art. Exemplary range of technical and wider ranging sources used and discussed in depth, indicating broad and critical background reading.*	*Very good evaluation of the state of the art. An appropriate range of relevant sources used and evaluated, indicating substantial background reading and consideration of the wider context of the problem.*	*Sufficient evaluation of the state of the art and formulation of the problem sufficiently supported by relevant and trustworthy references.*	*Satisfactory evaluation of the state of the art and formulation of the problem by few references. Typically, weak referencing and some questionable sources. .*	*Too few relevant references used and discussed, limited to technical area, indicating insufficient wider reading. Reliance on doubtful sources, resulting in a weak motivation and problem statement.*	*Only a few references used and discussed, and/or many are irrelevant. Little evidence of evaluation and problem statement.*	*Very few (or no) references used or discussed. No evidence of any evaluation and resulting problem statement.*
Technical / Design Narrative *(Weighting = 3)*	*Authoritative account of the novel solution of a complex problem, supported by critical evaluation at each stage of the design / technical analysis.*	*A lucid, coherent narrative, dictated by significant analyses, indicates a very good grasp and novel solution of a difficult technical problem.*	*The narrative is clear and shows how good technical or design choices followed from key technical analyses.*	*The narrative is of reasonable technical depth, indicates how analyses informed project direction, and shows satisfactory understanding.*	*Limited explanation of the technical work / design choices. Little or trivial technical analysis. Shortfalls of understanding in key areas.*	*Muddled discussion of technical work or results. Superficial understanding of the technical / design problem.*	*The lack of quality of the technical narrative suggests that the student has no real understanding of the problem.*
Discussion and Conclusion *(Weighting = 1)*	*Professional conclusion presenting key technical results with notable insights as to their implications for engineering / society. Discussion comparing the results to the state-of-the-art expertly backed up by relevant and trustworthy references.*	*Conclusion presents key technical results, critically assessing the wider relevance to current and future societal needs. Discussion compares the results to the state-of-the-art backed up by relevant and trustworthy references.*	*Minor omissions in the conclusion presenting the technical results and their relation to the wider engineering and societal context. Discussion compares the results backed up by mostly relevant references.*	*Conclusion shows a satisfactory understanding of the technical results and notes wider engineering and societal context. Discussion compares the results backed up by relevant but small number of references.*	*Conclusion shows a weak understanding of the technical results and wider engineering implications, and societal impact. Discussion has major flaws evaluating the work against the state of the art and/or uses dubious references.*	*Conclusion is mostly disconnected from the technical results and not relating them to the wider context. Discussion does not compare the work against the state of the art with no or dubious references.*	*No attempt at any point in the text to draw conclusions, discuss the work against the state of the art or put the work into a wider context.*

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(Academic Year 2024-25 onwards)

3. Poster Presentation (worth 15%)

(To be completed by James Watt School of Engineering staff only.)

Grade Range (Highest to Lowest)	**A1, A2, A3, A4, A5**	**B1, B2, B3**	**C1, C2, C3**	**D1, D2, D3**	**E1, E2, E3**	**F1, F2, F3**	**G1, G2, H**
Descriptor	**Excellent**	**Very Good**	**Good**	**Satisfactory**	**Weak**	**Poor**	**G: Very Poor** **H: No Attainment**
Introductory Description *(Weighting = 1)*	*Extremely clear and well-presented description of the project. Easy to follow arguments.*	*Very clear description of the project but perhaps a few minor flaws (e.g. hesitation, talking too fast).*	*Clear description of the project but some flaws in presentation. Not speaking quite clearly enough.*	*Overall a reasonable description, but there are issues regarding clarity or fluency.*	*Unclear or hesitant description made understanding the purpose of the project difficult.*	*Hesitant, unclear, monotonous, hard to maintain attention. Difficult to follow arguments.*	*No fluency or clarity. Too many basic errors. Project description too difficult to follow.*
Presentation of Content *(Weighting = 1)*	*Exceptionally clear presentation of poster content. Simple illustrations, large enough font, not too many words. A professional poster.*	*Clear presentation but perhaps the occasional flaw (font size, colour scheme etc), but overall an impressive poster.*	*There may be a number of errors in presentation but overall, still clear and flaws do not detract significantly from content.*	*Consistent errors in presentation but not of a significant nature. A reasonable effort but flaws detract.*	*Significantly flawed poster. Basic errors such as small font size, too much content on poster, illustrations too confusing or unclear.*	*Not only is the presentation of poster poor, but it makes it difficult to follow argument.*	*Very poor poster, basic errors in presentation throughout. Impossible to follow the technical arguments.*
Poster Layout *(Weighting = 1)*	*Structure of the poster makes understanding the technical arguments exceptionally clear.*	*A very well-structured poster with everything where it should be to provide clarity.*	*Overall a well-structured poster but perhaps one or two items are in the wrong position.*	*Some elements of the poster are not clear as the structure is slightly confused.*	*A badly structured poster giving a confused picture of the project making it difficult to follow the arguments.*	*Although there is some structure to the poster it is very confused, and it is almost impossible to follow.*	*No discernible attempt at a logical poster layout.*
Technical Content *(Weighting = 2)*	*There is a well-judged amount of high-level technical content on the poster giving an excellent account of a challenging technical task.*	*The poster has a very good level of technical content, clearly expressed, with only a small amount of superfluous information.*	*Overall, the content is sufficient to give the viewer a good account of the technical work undertaken.*	*There is some irrelevant non-pertinent material, but overall, the technical content is satisfactory.*	*The poster has only limited technical content with too much general background information.*	*The technical content is low in terms of level and quantity.*	*Little or no relevant technical content evident.*
Response to Questions *(Weighting = 2)*	*Answered all questions clearly and confidently. Gave the impression of having an excellent grasp of the subject.*	*Answered all questions competently. Has clearly developed a very good understanding of the subject.*	*Answered most questions well enough to conclude that the student has a developed a good understanding of the subject.*	*Gave some good answers but also some poor ones. Evidence of reasonable understanding of the subject.*	*Answered the majority of the questions poorly suggesting a lack of knowledge in the subject.*	*Gave some superficial answers but appears to have very little understanding of the subject.*	*Unable to give any sort of competent answer to any question.*